1. Display the current logged-in username, UID, primary GID and supplementary groups.

```
akansha@localhost:~$ id
uid=1000(akansha) gid=1000(akansha) groups=1000(akansha),10(wheel)
akansha@localhost:~$ whoami
akansha
```

2. Display the account entry of a test user from /etc/passwd and identify the fields stored in the entry.

```
akansha@localhost:~$ grep "^akansha:" /etc/passwd
akansha:x:1000:1000:Akansha:/home/akansha:/bin/bash
```

3. Display the password/account status of a test user without displaying the actual password.

```
akansha@localhost:~$ sudo passwd -S akansha
akansha PS 2026-10-31 0 99999 7 -1 (Password set, SHA512 crypt.)
```

4. Create a new user named labuser1 using useradd and verify that the account has been created.

```
akansha@localhost:~$ sudo useradd labuser1
akansha@localhost:~$ grep "^labuser1:" /etc/passwd
labuser1:x:1001:1001::/home/labuser1:/bin/bash
```

5. Create labuser2 with a home directory and verify the home directory and its ownership.

```
akansha@localhost:~$ sudo useradd -m labuser2
akansha@localhost:~$ ls -ld /home/labuser2
drwx------. 2 labuser2 labuser2 6 Oct 31 11:05 /home/labuser2
```

6. Create labuser3 with /bin/bash as the login shell and verify the assigned shell.

```
akansha@localhost:~$ sudo useradd -s /bin/bash labuser3
akansha@localhost:~$ grep "^labuser3:" /etc/passwd | cut -d: -f7
/bin/bash
```

7. Set a password for labuser1 and verify the password/account status.

```
akansha@localhost:~$ sudo passwd labuser1
Changing password for user labuser1.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
akansha@localhost:~$ sudo passwd -S labuser1
labuser1 PS 2026-10-31 0 99999 7 -1
```

8. Change the password of labuser1 and verify the account status again.

```
akansha@localhost:~$ sudo passwd labuser1
Changing password for user labuser1.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
akansha@localhost:~$ sudo passwd -S labuser1
labuser1 PS 2026-10-31 0 99999 7 -1
```

9. Use usermod to change labuser2's login shell to /bin/bash and verify the change.

```
akansha@localhost:~$ sudo usermod -s /bin/bash labuser2
akansha@localhost:~$ grep "^labuser2:" /etc/passwd
labuser2:x:1002:1002::/home/labuser2:/bin/bash
```

10. Use usermod to change labuser2's home directory, first without moving existing contents and then with the option that moves contents.

```
akansha@localhost:~$ sudo usermod -d /home/labuser2_new labuser2
akansha@localhost:~$ grep "^labuser2:" /etc/passwd
labuser2:x:1002:1002::/home/labuser2_new:/bin/bash
akansha@localhost:~$ sudo usermod -m -d /home/labuser2_moved labuser2
akansha@localhost:~$ ls -ld /home/labuser2_moved
drwx------. 2 labuser2 labuser2 6 Oct 31 11:05 /home/labuser2_moved
```

11. Create a supplementary group named userlabgroup and add labuser1 to it using usermod. Verify membership.

```
akansha@localhost:~$ sudo groupadd userlabgroup
akansha@localhost:~$ sudo usermod -aG userlabgroup labuser1
akansha@localhost:~$ id labuser1
uid=1001(labuser1) gid=1001(labuser1) groups=1001(labuser1),1004(userlabgroup)
```

12. Lock labuser3 and verify that the account is locked. Then unlock it and verify the result.

```
akansha@localhost:~$ sudo usermod -L labuser3
akansha@localhost:~$ sudo passwd -S labuser3
labuser3 LK 2026-10-31 0 99999 7 -1 (Password locked.)
akansha@localhost:~$ sudo usermod -U labuser3
akansha@localhost:~$ sudo passwd -S labuser3
labuser3 NP 2026-10-31 0 99999 7 -1
```

13. Display detailed password-aging information for labuser3 using chage.

```
akansha@localhost:~$ sudo chage -l labuser3
Last password change                                    : Oct 31, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
```

14. Set an account expiration date for labuser3 and verify the expiration information.

```
akansha@localhost:~$ sudo chage -E 2026-12-31 labuser3
akansha@localhost:~$ sudo chage -l labuser3 | grep "Account expires"
Account expires                                         : Dec 31, 2026
```

15. Modify labuser1 so that a specified existing group becomes the user's primary group. Verify result.

```
akansha@localhost:~$ sudo usermod -g userlabgroup labuser1
akansha@localhost:~$ id labuser1
uid=1001(labuser1) gid=1004(userlabgroup) groups=1004(userlabgroup)
```

16. Display users and groups associated with a selected account using id, groups and getent.

```
akansha@localhost:~$ id labuser1
uid=1001(labuser1) gid=1004(userlabgroup) groups=1004(userlabgroup)
akansha@localhost:~$ groups labuser1
labuser1 : userlabgroup
akansha@localhost:~$ getent passwd labuser1
labuser1:x:1001:1004::/home/labuser1:/bin/bash
```

17. Delete labuser1 without removing its home directory. Verify account entry and home directory separately.

```
akansha@localhost:~$ sudo userdel labuser1
akansha@localhost:~$ grep "^labuser1:" /etc/passwd
akansha@localhost:~$ ls -ld /home/labuser1
drwx------. 2 1001 1004 6 Oct 31 11:00 /home/labuser1
```

18. Delete labuser2 and remove its home directory using appropriate userdel option. Verify result.

```
akansha@localhost:~$ sudo userdel -r labuser2
akansha@localhost:~$ grep "^labuser2:" /etc/passwd
akansha@localhost:~$ ls -ld /home/labuser2_moved
ls: cannot access '/home/labuser2_moved': No such file or directory
```

19. Create a complete user-management workflow: create a user, set a password, assign a shell, add a supplementary group, lock/unlock account and verify all settings.

```
akansha@localhost:~$ sudo useradd -m -s /bin/bash testuser
akansha@localhost:~$ echo "testuser:Password123" | sudo chpasswd
akansha@localhost:~$ sudo groupadd testgroup
akansha@localhost:~$ sudo usermod -aG testgroup testuser
akansha@localhost:~$ sudo usermod -L testuser
akansha@localhost:~$ sudo passwd -S testuser
testuser LK 2026-10-31 0 99999 7 -1
akansha@localhost:~$ sudo usermod -U testuser
akansha@localhost:~$ id testuser
uid=1004(testuser) gid=1005(testuser) groups=1005(testuser),1006(testgroup)
```

20. Clean up all test users and lab-only groups created during this lab. Verify that no unintended system account has been modified.

```
akansha@localhost:~$ sudo userdel -r labuser3 2>/dev/null
akansha@localhost:~$ sudo userdel -r testuser 2>/dev/null
akansha@localhost:~$ sudo groupdel userlabgroup 2>/dev/null
akansha@localhost:~$ sudo groupdel testgroup 2>/dev/null
akansha@localhost:~$ sudo rm -rf /home/labuser1
akansha@localhost:~$ getent passwd | tail -n 5
akansha:x:1000:1000:Akansha:/home/akansha:/bin/bash
```